

A+ Core 2 Memorization Sheet

Print this and keep it beside you. Core 2 is about names: which Windows tool, which command, which edition, which security term, which step comes next, which symptom points where. Every line here is fair game on the exam.

01 The exam itself

FACT	VALUE
Exam	CompTIA A+ Core 2, 220-1202 (A+ needs Core 1 220-1201 and Core 2)
Questions and time	Up to 90 questions in 90 minutes; multiple choice plus performance-based items
Passing	700 on a scale of 100 to 900 (Core 1 needs 675)
Operating Systems	28%
Security	28%
Software Troubleshooting	23%
Operational Procedures	21%

Operating systems and security are more than half the exam. Know every tool by its file name, every command by its job, and every procedure in its exact order.

02 Operating systems, filesystems, and Windows editions

OPERATING SYSTEM	FACTS
Windows	Workstation OS; Home, Pro, Pro for Workstations, Enterprise; NTFS; domain or workgroup; the only one with Group Policy and BitLocker
Linux	Open-source kernel with distributions (Ubuntu and Debian use apt, Fedora and RHEL use dnf); ext4 or XFS; case-sensitive; root account; systemd
macOS	Apple desktops and laptops only; APFS; App Store; Time Machine; FileVault; Keychain; Spotlight
Chrome OS	Google's browser-centered OS on Chromebooks; cloud storage, Android apps, managed from the Google Admin console
iOS and iPadOS	Apple phones and tablets; closed app store; no sideloading without enterprise profiles
Android	Google's open mobile OS; many vendors; sideloading possible; more fragmentation of updates
Vendor life cycle	End-of-life (EOL): no more security updates; update limitations: hardware or edition blocks a version; Windows 10 support ended October 2025
Compatibility	Applications and drivers are built per OS and per architecture; 32-bit apps run on 64-bit Windows, not the reverse; files move between OSes only on filesystems both can read (exFAT, FAT32)

FILESYSTEM	WHERE	REMEMBER
NTFS	Windows system and data drives	Permissions, encryption (EFS), compression, journaling, huge volumes and files
ReFS	Windows Server and Pro for Workstations	Resilient: integrity streams, self-healing, very large volumes; not bootable in most editions
FAT32	Old drives, small flash, firmware	Universal compatibility; 4 GB maximum file size; Windows formats only up to 32 GB

FILESYSTEM	WHERE	REMEMBER
exFAT	Large flash drives and SD cards	Big files, no 4 GB limit, readable by Windows, macOS, and Linux; no permissions or journaling
ext4	Linux default	Journaling, large files; Windows cannot read it without extra software
XFS	Linux (RHEL default), servers	High performance with very large files and volumes
APFS	macOS and iOS	Snapshots, encryption, cloning, SSD optimized; replaced HFS+

WINDOWS EDITION	WHAT IT ADDS
Home (10 and 11)	Consumer; workgroup only, no domain join; no BitLocker, no Remote Desktop host, no Group Policy Editor (gpedit.msc); 128 GB RAM limit
Pro (10 and 11)	Domain join, BitLocker, Remote Desktop host, gpedit.msc, Hyper-V; 2 TB RAM
Pro for Workstations (10)	ReFS, persistent memory, up to four CPUs, 6 TB RAM; for high-end workstations
Enterprise (10 and 11)	Volume licensing; adds AppLocker, DirectAccess, advanced management; 6 TB RAM
N versions	European editions without Media Player and media features
Desktop and RDP	Windows 11 has the centered taskbar and new Settings; every edition can be an RDP client, only Pro and above can be an RDP host
Upgrade paths	In-place upgrade keeps files, apps, and settings (Windows 10 to 11 if the hardware qualifies; Home to Pro with a key); clean install wipes; 32-bit to 64-bit requires a clean install
Windows 11 hardware	TPM 2.0, UEFI with Secure Boot capable, 64-bit dual-core 1 GHz CPU, 4 GB RAM, 64 GB storage

03 Installation, upgrades, and application requirements

BOOT OR INSTALL ITEM	FACTS
Boot methods	USB, network (PXE, Preboot Execution Environment), solid-state or flash drive, internet-based (cloud recovery), external or hot-swappable drive, internal drive partition, multiboot (boot menu chooses the OS)
Clean install	Wipes the drive; new OS; restore files afterward
Upgrade (in-place)	Keeps files, apps, and settings; needs a supported path and compatible hardware and drivers
Image deployment	A prepared disk image copied to many machines (cloning, MDT, Configuration Manager)
Remote network installation	Boot from PXE, install from a server share
Zero-touch deployment	Device ships to the user and enrolls itself through the vendor's service (Windows Autopilot); no technician touches it
Recovery partition	Vendor partition that restores the factory image
Repair installation	Reinstalls Windows over itself, keeping data and apps, to fix system files
Third-party drivers	Load storage or RAID drivers during setup when the installer cannot see the disk
GPT	GUID partition table: UEFI, up to 128 partitions, disks over 2 TB
MBR	Master boot record: legacy BIOS, four primary partitions (or three plus an extended), 2 TB limit

BOOT OR INSTALL ITEM	FACTS
Upgrade considerations	Back up files and user preferences; check application and driver support and backward compatibility; hardware compatibility; feature updates twice a year become once a year; product life cycle (EOL date)

Application requirements and impact

- 32-bit versus 64-bit: a 64-bit app needs a 64-bit OS; a 32-bit app runs on either; 32-bit is limited to about 4 GB of addressable memory
- Dedicated graphics card with its own VRAM for gaming, CAD, and video; integrated graphics shares system RAM; check VRAM, RAM, CPU, and storage minimums against the machine, not just the OS
- External hardware tokens: some software licenses require a USB dongle or key present to run
- Distribution: physical media, a mountable ISO file, a downloadable package, or image deployment; verify the download's hash
- Impact considerations: device (resources, compatibility), network (bandwidth, ports, licensing servers), operation (training, downtime, workflows), business (cost, compliance, support)
- Cloud productivity: email systems, storage with sync and folder settings, collaboration tools (spreadsheets, videoconferencing, presentations, word processing, instant messaging), identity synchronization with the directory, licensing assignment per user

04 Windows tools by file name

TOOL	OPEN WITH	USE IT FOR
Task Manager	Ctrl+Shift+Esc; taskmgr	Processes (end a hung app, see CPU, memory, disk, network), Performance graphs, Startup apps, Users, Services
Event Viewer	eventvwr.msc	Application, Security, System logs; errors and warnings behind crashes and failed services
Disk Management	diskmgmt.msc	Initialize, partition, format, extend, shrink, assign letters, convert MBR and GPT, dynamic disks
Task Scheduler	taskschd.msc	Run scripts and programs at a time or on an event
Device Manager	devmgmt.msc	Drivers: update, roll back, disable; yellow triangle means a problem; resource conflicts
Certificate Manager	certmgr.msc	View, import, and remove user certificates; trusted roots
Local Users and Groups	lusrmgr.msc	Create accounts, reset passwords, group membership (Pro and above)
Performance Monitor	perfmon.msc	Counters over time; data collector sets; baselines
Group Policy Editor	gpedit.msc	Local policies for the computer and users (Pro and above)
System Information	msinfo32.exe	Hardware, drivers, software inventory; BIOS mode; export a report
Resource Monitor	resmon.exe	Live CPU, memory, disk, network per process; find what holds a file
System Configuration	msconfig.exe	Boot options (safe boot), services, startup (now in Task Manager), tools
Disk Cleanup	cleanmgr.exe	Remove temporary files, old updates, previous installations
Disk Defragmenter	dfrgui.exe	Optimize drives: defragment HDDs, TRIM SSDs; scheduled weekly
Registry Editor	regedit.exe	Edit the registry hives (HKLM, HKCU); export a backup before changing anything
Microsoft Management Console	mmc.exe	Host for snap-ins; build a custom console

05 Command-line tools

WINDOWS COMMAND	JOB
cd, dir	Change directory; list files (dir /a shows hidden)
md, rmdir	Make a directory; remove a directory (rmdir /s removes contents)
robocopy	Robust copy of folder trees with retries, mirroring (/mir), and permissions; xcopy is the old one
ipconfig	Address, mask, gateway; /all adds MAC and DNS; /release and /renew for DHCP; /flushdns clears the resolver cache
ping	Reachability and round trip; -t continuous; ping 127.0.0.1 tests the stack
tracert	Every hop to a destination
pathping	Tracert plus per-hop packet loss statistics over time
netstat	Open connections and listening ports; -a all, -n numeric, -b program names
nslookup	Query DNS for a name or address
net use	Map a drive: net use X: \\server\share; net use * /delete removes mappings
net user	List or manage local accounts; net user name password /add
chkdsk	Check the disk; /f fixes errors; /r locates bad sectors and recovers data
format	Format a volume with a filesystem; destroys data
diskpart	Command-line partitioning: list disk, select disk, clean, create partition, convert gpt
hostname, whoami, winver	Computer name; current user and groups; Windows version dialog
[command] /?	Help for any command
gpupdate	Apply Group Policy now; /force reapplies everything
gpresult	Show applied policies; /r summary
sfc	System File Checker; sfc /scannow repairs protected system files (DISM /online /cleanup-image /restorehealth repairs the source)
shutdown	shutdown /r /t 0 restarts now; /s shuts down
LINUX COMMAND	JOB
ls, pwd	List files (ls -la shows all with permissions); print the working directory
cd, mv, cp, rm	Change directory; move or rename; copy (cp -r for folders); remove (rm -rf removes a tree; no undo)
chmod, chown	Change permissions (r 4, w 2, x 1; owner group other, so 755 and 644); change owner and group
grep, find	Search text inside files; find files by name, size, or age
cat, man	Print a file; read the manual page for a command
top, ps	Live process view; list processes (ps aux)
df, du	Free space per filesystem; space used by a directory (du -sh)
fsck, mount	Check a filesystem (unmounted); attach a filesystem to a directory (umount detaches)
su, sudo	Switch user (to root with a password); run one command as root (sudo, logged, preferred)
apt, dnf	Package managers: apt on Debian and Ubuntu (apt update, apt install); dnf on Fedora and RHEL
ip, ping, curl, dig, traceroute	ip addr and ip route for addressing; ping; curl fetches URLs; dig queries DNS; traceroute lists hops
nano	Simple terminal text editor (vi and vim are the others)

LINUX COMMAND	JOB
<code>/etc/passwd, /etc/shadow</code>	User accounts; hashed passwords (root only)
<code>/etc/hosts, /etc/fstab, /etc/resolv.conf</code>	Static name mappings; filesystems to mount at boot; DNS servers
<code>systemd, kernel, bootloader</code>	Service manager (<code>systemctl start, enable, status</code>); the kernel is the core; GRUB is the bootloader
<code>root</code>	The all-powerful account (UID 0); log in as a normal user and use <code>sudo</code>

06 Control Panel, Settings, and Windows networking

CONTROL PANEL APPLET	WHAT LIVES THERE
Internet Options	Browser security zones, proxy (Connections tab, LAN settings), certificates, home page for legacy Internet Explorer settings
Devices and Printers	Add, remove, set default, printer properties and queues
Programs and Features	Uninstall or change programs; turn Windows features on or off; view installed updates
Network and Sharing Center	Adapter settings, network profile, sharing options, troubleshooting
System	Computer name, domain or workgroup, remote settings, System Protection (restore points), advanced (performance, virtual memory, environment variables)
Windows Defender Firewall	Turn on or off per profile; allow an app; advanced inbound and outbound rules by port
Mail	Outlook profiles and data files
Sound	Playback and recording devices, default device, levels
User Accounts	Change account type, passwords, UAC level, credential manager
Device Manager	The same <code>devmgmt.msc</code>
Indexing Options	What Windows Search indexes and where
Administrative Tools	Shortcuts to the MMC consoles
File Explorer Options	General (open in This PC), View (show hidden files, show file extensions, protected OS files), Search
Power Options	Power plans (Balanced, Power saver, High performance); sleep and hibernate timers; what closing the lid does; fast startup; USB selective suspend; standby; hibernate writes RAM to disk, sleep keeps RAM powered

SETTINGS CATEGORY	WHAT LIVES THERE
System	Display, sound, notifications, power and sleep, storage, about
Devices	Bluetooth, printers, mouse, typing, pen, AutoPlay, USB
Network and Internet	Status, Wi-Fi, Ethernet, VPN, proxy, data usage, metered connections
Personalization	Background, colors, lock screen, themes, start, taskbar
Apps	Apps and features (uninstall), default apps, optional features, startup
Accounts	Your info, email accounts, sign-in options (PIN, Windows Hello), access work or school, family
Time and Language	Date and time, region, language, speech
Gaming	Game bar, captures, game mode

SETTINGS CATEGORY	WHAT LIVES THERE
Ease of Access	Accessibility: narrator, magnifier, high contrast, captions
Privacy	Location, camera, microphone, diagnostics, app permissions
Update and Security	Windows Update, delivery optimization, Windows Security, backup, troubleshoot, recovery, activation, Find My Device

Windows networking

- Domain joined: central accounts and Group Policy from Active Directory; workgroup: local accounts on each PC, small networks
- Shared resources: printers, file servers, mapped drives (net use or Map network drive); UNC paths \\server\share in File Explorer
- Local firewall: allow an application through, add port exceptions, separate rules per profile; public network profile blocks discovery and sharing, private allows it
- Client configuration: IP address, subnet mask, gateway, DNS servers; static (typed) or dynamic (DHCP); alternate configuration
- Connections: VPN (built-in client, add a VPN connection), wireless (SSID and security type), wired, WWAN cellular (SIM, APN, data plan)
- Proxy settings: automatic detection, setup script, or manual address and port; metered connections limit updates and background data

07 macOS and Linux features

MACOS ITEM	FACTS
File types	.dmg is a disk image you mount and drag the app from; .pkg is an installer package; .app is the application bundle itself; App Store installs and updates; uninstall by dragging to Trash or running the vendor uninstaller
System folders	/Applications (apps), /Users (home folders), /Library (system-wide support files), /System (the OS, protected), /Users/name/Library (per-user settings, hidden)
Apple ID and corporate restrictions	Personal Apple ID for iCloud and the App Store; managed Apple IDs and MDM restrict what corporate Macs can do
Best practices	Backups with Time Machine, antivirus, updates and patches, Rapid Security Response (small security patches between full updates)
System Preferences / Settings	Displays, Network, Printers and Scanners, Privacy and Security, Accessibility, Time Machine
Features	Multiple desktops (Spaces) and Mission Control, Keychain (passwords), Spotlight (search), iCloud, iMessage, FaceTime, iCloud Drive, trackpad gestures, Finder (file manager), Dock, Continuity and Handoff between Apple devices
Utilities	Disk Utility (partition, format, First Aid), FileVault (full-disk encryption), Terminal (shell), Force Quit (Command-Option-Esc)

08 Security concepts

Physical security and physical access

- Bollards stop vehicles; access control vestibule (mantrap) admits one person at a time; badge reader; video surveillance; alarm systems; motion sensors; door locks; equipment locks (cable locks); security guards; fences; lighting; magnetometers (metal detectors)
- Physical access: key fobs, smart cards, mobile digital keys, metal keys; biometrics: retina, fingerprint, palm print, facial recognition, voice recognition

LOGICAL SECURITY TERM	MEANING
Principle of least privilege	Every account gets only the rights its job needs
Zero Trust	Never trust by location; verify every user and device for every request
Access control list	The list of who may do what to a resource (files, firewalls, routers)
MFA methods	Email code, hardware token, authenticator application, SMS, voice call, TOTP (time-based one-time password), OTP; SMS and voice are the weakest
SAML	Security Assertion Markup Language: passes authentication between an identity provider and a web application; the basis of web SSO
SSO	One login for many systems
Just-in-time access	Elevated rights granted only for the task and the time needed
PAM	Privileged access management: vaulted admin credentials, checked out, logged
MDM	Mobile device management: enroll, configure, enforce, wipe
DLP	Data loss prevention: stops sensitive data leaving by email, upload, or USB
IAM	Identity and access management: the whole system of accounts, roles, and authentication
Directory services	Active Directory and LDAP: the database of users, computers, and groups
WIRELESS SECURITY	FACTS
WPA2	AES-CCMP encryption; personal (pre-shared key) or enterprise (802.1X with RADIUS); the minimum acceptable today
WPA3	Stronger: SAE handshake resists offline password guessing, forward secrecy, protected management frames; enterprise 192-bit mode
TKIP	The old WPA encryption; deprecated, weak; never choose it
AES	Advanced Encryption Standard: the strong cipher in WPA2 and WPA3
RADIUS	Central authentication for Wi-Fi and VPN; UDP 1812 and 1813; encrypts only the password
TACACS+	Cisco device administration authentication; TCP 49; encrypts the whole session; separates authentication, authorization, accounting
Kerberos	Active Directory's ticket-based authentication; the KDC issues tickets; clocks must agree within about five minutes
Multifactor	Something you know, have, or are; two of them
MALWARE	SIGNATURE
Trojan	Pretends to be useful software; installs a backdoor
Rootkit	Hides deep in the OS or firmware; invisible to normal tools; often needs a reinstall
Virus	Attaches to files; spreads when they run
Spyware	Watches and reports activity
Ransomware	Encrypts files and demands payment
Keylogger	Records keystrokes for passwords
Boot sector virus	Infects the boot record; loads before the OS
Cryptominer	Steals CPU and GPU to mine currency; slow and hot machine

MALWARE	SIGNATURE
Stalkerware	Secretly tracks a person's location and messages
Fileless	Lives in memory and legitimate tools (PowerShell); no file for antivirus to scan
Adware and PUP	Ads and unwanted bundled programs; nuisance more than threat
Tools	Recovery console (WinRE), EDR, MDR (managed by a provider), XDR (across endpoints and network), antivirus, antimalware, email security gateway, software firewalls, user education and antiphishing training, OS reinstallation as the last resort
SOCIAL ENGINEERING AND THREATS	SIGNATURE
Phishing, vishing, smishing	Fake email, voice call, SMS; QR code phishing hides the link in a code
Spear phishing and whaling	Targeted at a person; targeted at an executive
Shoulder surfing, tailgating	Watching a screen or keyboard; following someone through a door
Impersonation, dumpster diving	Pretending to be someone; digging paper out of the trash
DoS and DDoS	Flooding a service; distributed from many bots
Evil twin	A rogue access point with the real network's name
Zero-day	Exploits a flaw with no patch yet
Spoofing, on-path	Faking an address or identity; intercepting between two parties
Brute force, dictionary	Trying every password; trying likely words
Insider threat	An authorized person misusing access
SQL injection, XSS	Database commands through a form; scripts injected into a web page
BEC, supply chain	Business email compromise (fake invoice from the boss); attack through a vendor or software pipeline
Vulnerabilities	Non-compliant, unpatched, unprotected (no antivirus or firewall), EOL systems, BYOD devices

09 Windows security, Active Directory, hardening, and mobile security

WINDOWS SECURITY SETTING	FACTS
Defender Antivirus	Built in; turn on, keep definitions updated; real-time protection
Firewall	On per profile; port rules and application rules; block inbound by default
Accounts	Local versus Microsoft account; Standard user; Administrator; Guest (disabled); Power Users (legacy group)
Sign-in options	Username and password, PIN, fingerprint, facial recognition, SSO, passwordless Windows Hello
NTFS versus share permissions	Share permissions apply only over the network; NTFS applies locally and remotely; the effective permission is the most restrictive of the two; an explicit deny wins
NTFS rules	Copying to another volume takes the destination's permissions; moving within a volume keeps the original; inheritance flows from the parent unless blocked; attributes: read-only, hidden, system, archive
UAC	User Account Control prompts before elevation; run as administrator only when needed; standard users enter admin credentials
BitLocker and To Go	Full-volume encryption keyed to the TPM (Pro and above); BitLocker To Go for removable drives; save the recovery key
EFS	Encrypting File System: per-file or per-folder encryption tied to the user's certificate; lose the key, lose the files

Active Directory tasks

- Join the domain from System settings; the computer object appears in AD; users then log in with domain accounts
- Login scripts run at sign-in to map drives and printers; home folders give each user a server folder; folder redirection points Documents and Desktop to the server for backup
- Organizational units group users and computers so Group Policy can be applied per OU; move objects between OUs to change what applies
- Security groups hold users for permissions; assign permissions to groups, never to individuals; Group Policy enforces settings across the domain

Workstation hardening

- Data-at-rest encryption (BitLocker, FileVault); BIOS and UEFI passwords
- Passwords: length first, then character types, uniqueness, complexity, expiration; password managers instead of reuse
- Users: screensaver locks, log off when away, secure laptops with cable locks, protect PII and passwords
- Accounts: restrict permissions, restrict login times, disable the guest account, failed-attempt lockout, timeout and screen lock, account expiration dates
- Change the default administrator account and password; disable AutoRun and AutoPlay; disable unused services

Mobile device security

- Device encryption; screen locks (facial recognition, PIN, fingerprint, pattern, swipe; swipe is not security); configuration profiles pushed by MDM
- Patch management for the OS and apps; endpoint security (antivirus, antimalware, content filtering)
- Locator applications, remote wipe, remote backup, failed-login restrictions (wipe after N attempts)
- Policies: MDM enrollment, BYOD versus corporate-owned, profile security requirements

10 SOHO, browser, disposal, and the malware removal procedure

SOHO router security

- Change default passwords; update firmware; secure management access (HTTPS, disable remote or WAN management); physical placement in a secure spot
- IP filtering and MAC filtering; content filtering; disable UPnP; screened subnet (DMZ) for exposed servers
- Wireless: change the SSID, disabling SSID broadcast hides but does not secure, WPA3 or WPA2-AES, guest network isolated from the LAN
- Firewall: disable unused ports, port forwarding or mapping only for needed services

Browser security

- Download from trusted sources; verify hashes; avoid untrusted sources; keep the browser patched
- Extensions and plug-ins only from trusted stores; review permissions; remove unknown ones
- Password managers; valid certificates (padlock, no warnings); pop-up blocker; clear browsing data and cache; private browsing mode; sign-in and sync across devices; ad blockers; proxy; secure DNS (DNS over HTTPS)
- Browser feature management: enable or disable plug-ins, extensions, and features by policy

DISPOSAL METHOD	FACTS
Drilling, shredding, incineration	Physical destruction; certain; nothing recoverable
Degaussing	A strong magnet erases magnetic drives and tape; useless on SSDs
Erasing and wiping	Overwriting or secure erase; keeps the drive usable for reuse
Low-level format	Factory-level reset of the drive surface; modern drives do it in firmware
Standard format	Rebuilds the filesystem; a quick format leaves data recoverable
Outsourcing	Third-party vendor with a certificate of destruction or recycling

DISPOSAL METHOD	FACTS
Regulations	E-waste rules and data laws decide what must be destroyed and how

Malware removal, in order

- 1 Investigate and verify malware symptoms. 2 Quarantine the infected system (disconnect it from the network). 3 Disable System Restore in Windows Home. 4 Remediate: update the antimalware software, scan and remove (safe mode, preinstallation environment), reimaged or reinstall if needed. 5 Schedule scans and run updates. 6 Enable System Restore and create a restore point in Windows Home. 7 Educate the end user.

11 Software troubleshooting

WINDOWS SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Blue screen of death	Driver, hardware, or memory fault; note the stop code; recent changes; safe mode; roll back the driver; memory test
Degraded performance	Too many startup items, low RAM, full or failing disk, malware, background updates; Task Manager and Resource Monitor
Boot issues	Corrupted boot files or wrong boot order; Startup Repair in WinRE; bootrec; check the drive
Frequent shutdowns	Overheating, failing power supply, driver; check temperatures and Event Viewer
Services not starting	Dependency or account problem; Services console, Event Viewer, sfc
Applications crashing	Update or reinstall the app; check Event Viewer; compatibility mode; sfc
Low memory warnings	Not enough RAM or paging file; close apps; add RAM; check for leaks
USB controller resource warnings	Too many devices on one controller; move devices to another port or a powered hub
System instability	Bad drivers, failing hardware, corrupt system files; sfc, DISM, memory test, roll back updates
No OS found	Boot order, unplugged drive, corrupted boot sector, USB stick left in; firmware settings; Startup Repair
Slow profile load	Large profile, roaming profile, login scripts, mapped drives timing out, corrupt profile (rebuild it)
Time drift	CMOS battery or a time source failure; sync with the domain or NTP; certificates and Kerberos fail when time is off

Windows repair toolbox

- Safe mode (minimal drivers), WinRE (Startup Repair, System Restore, Command Prompt, Reset this PC), bootrec /fixmbr /fixboot /rebuildbcd, sfc /scannow, DISM, chkdsk /r, memory diagnostic, driver rollback, uninstall updates, rebuild the user profile, reinstall or repair install

MOBILE SYMPTOM	LIKELY CAUSE AND FIRST CHECK
App fails to launch, close, or crashes	Force stop, clear cache, update, reinstall, restart the device
App fails to update or install	Storage full, OS version too old, no network, store account problem, MDM restriction
Slow to respond	Storage full, too many background apps, old OS, overheating, malware; restart
OS fails to update	Storage, battery level, network, unsupported (EOL) device
Battery life issues	Screen brightness, background apps, radios, poor signal, aged battery, malware
Random reboots	Bad app, OS bug, overheating, failing battery; update, remove recent apps, factory reset last

MOBILE SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Connectivity issues	Bluetooth: pair again; Wi-Fi: forget and rejoin, airplane mode; NFC: enable it, remove the case
Screen does not autorotate	Rotation lock is on; then the sensor or a calibration

SECURITY SYMPTOM	MEANING
Mobile: unofficial app stores, developer mode, root or jailbreak	Bypass the platform's protections; malware and unsupported by MDM
Mobile: unauthorized or malicious app, application spoofing	A fake copy of a real app; remove it, change passwords
Mobile: high network traffic, degraded response, data-usage warnings, ads, fake security warnings, unexpected behavior, leaked files	Malware or a malicious app; remove, scan, reset if needed
PC: cannot access the network, desktop alerts, false antivirus alerts, altered or missing or renamed files, cannot access files, unwanted notifications, update failures	Infection or a rogue security program; run the malware removal procedure
Browser: random pop-ups, certificate warnings, redirection, degraded performance	Adware, a hijacked browser, a bad extension, or an on-path attack; reset the browser, remove extensions, check DNS and hosts file

12 Documentation, change management, and backups

Documentation and support

- Ticket fields: user information, device information, description of the issue, category, severity, escalation level; write clearly: issue description, progress notes, resolution
- Asset management: inventory lists, configuration management database (CMDB), asset tags and IDs, procurement life cycle, warranty and licensing, assigned users
- Documents: incident reports, standard operating procedures (SOPs), custom software installation procedures, new-user onboarding checklist, user off-boarding checklist, service-level agreements (internal and external or third-party), knowledge base articles

Change management

- Documented business processes: rollback plan, backup plan, sandbox testing, responsible staff members
- Request form: purpose, scope, change type (standard is pre-approved and routine, normal goes to the board, emergency is fixed first and reviewed after), date and time, change freeze periods, maintenance windows
- Affected systems and impact, risk analysis and risk level, change board approvals, implementation, peer review, end-user acceptance

BACKUP TERM	MEANING
Full	Everything; slowest to make, fastest to restore
Incremental	Only what changed since the last backup of any kind; small and fast; restore needs the full plus every incremental in order
Differential	Everything changed since the last full; grows daily; restore needs the full plus the latest differential
Synthetic full	A full backup assembled on the backup server from the last full and later incrementals, without touching the client
Recovery	In-place or overwrite restores over the original; alternative location restores to a new folder or machine to compare
Testing	Restore regularly; a backup never tested is not a backup; set a frequency
Rotation	On-site (fast) versus off-site (disaster); grandfather-father-son: daily, weekly, monthly sets; 3-2-1 rule: three copies, two media types, one off-site

13 Safety, environment, policy, and professionalism

Safety

- ESD: wrist strap clipped to the chassis, ESD mat, antistatic bags, touch bare metal first; humidity around 40 to 60 percent reduces static; never wear a strap inside a power supply or CRT
- Electrical: equipment grounding; disconnect power before repairing a PC; do not open power supplies or CRTs (capacitors)
- Handling and storage: cable management, antistatic bags, hold boards by the edges; comply with government regulations
- Personal: lifting with the legs and asking for help on heavy items; fire safety (class C extinguisher for electrical; pull the plug first); safety goggles; air filter mask for dust and toner

Environment

- Safety data sheet (MSDS or SDS) for handling and disposal of chemicals, toner, batteries
- Disposal: batteries to a recycler (lithium is a fire hazard), toner cartridges returned or recycled, monitors and devices to e-waste; never the trash
- Temperature and humidity within the vendor's range; ventilation; placement away from heat, dust, and moisture; dust cleanup with compressed air outdoors or a vacuum rated for electronics or toner
- Power: surges, brownouts (sags), blackouts; a UPS bridges outages and conditions power; a surge suppressor clamps spikes (joule rating)

Prohibited content, incident response, licensing, and privacy

- Incident response: preserve the scene, document, chain of custody (who held the evidence, when), inform management and law enforcement as required, make a forensic copy of the drive (hash it for integrity), collect in order of volatility (CPU registers and cache, memory, swap and temporary files, disk, remote logs, archives)
- Licensing: valid licenses only; DRM enforces them; EULA is the contract; perpetual (buy once) versus subscription; personal-use versus corporate-use; open-source licenses allow use and modification under their terms
- NDA and mutual NDA protect confidential information both ways
- Regulated data: credit card payment information (PCI DSS), personal government-issued information, PII, healthcare data (HIPAA); data retention requirements set how long records are kept
- Acceptable use policy; regulatory and business compliance; splash screens (login banners) state the rules before access

Communication and professionalism

- Professional appearance matching the environment (formal or business casual); proper language without jargon, acronyms, or slang; positive attitude and confidence
- Actively listen, do not interrupt; be culturally sensitive; use professional titles; be on time, or call if late; avoid distractions (personal calls, texting, social media, interruptions)
- Difficult situations: do not argue or be defensive, do not dismiss or judge, clarify with open-ended questions and restate the issue, use discretion about what you discuss
- Set and meet expectations, communicate status, offer repair or replacement options, document the service, follow up later; handle confidential material on screens, desks, and printers with discretion

14 Scripting, remote access, and AI

SCRIPT TYPE	RUNS IN
.bat	Windows batch, cmd.exe; oldest and simplest
.ps1	PowerShell; object-based, the Windows administrator's tool; execution policy may block scripts
.vbs	VBScript; legacy Windows scripting (deprecated)
.sh	Shell (bash) on Linux and macOS
.js	JavaScript; browsers and Node

SCRIPT TYPE	RUNS IN
.py	Python; cross-platform automation
Use cases	Basic automation, restarting machines, remapping network drives, installing applications, automated backups, gathering information, initiating updates
Considerations	Scripts can unintentionally introduce malware, inadvertently change system settings, or crash a browser or system by mishandling resources; test in a sandbox, read before running
REMOTE ACCESS	FACTS
RDP	Remote Desktop Protocol, TCP 3389; full Windows desktop; host needs Pro or above; use with a VPN or gateway, never open to the internet
VPN	Encrypted tunnel to the office network; then use any tool inside
VNC	Cross-platform screen sharing, port 5900; weak security unless tunneled
SSH	Secure shell, TCP 22; encrypted command line for Linux, macOS, network gear
RMM	Remote monitoring and management agents used by service providers; patching, alerts, remote control
SPICE	Simple Protocol for Independent Computing Environments: remote display for virtual machines (KVM and QEMU)
WinRM	Windows Remote Management: PowerShell remoting over HTTP 5985 and HTTPS 5986
Third-party tools	Screen-sharing, videoconferencing, file transfer, desktop management software; check their security and who can connect
Security	Encryption, MFA, least privilege, logging, closing ports when done, verifying the person on the other end

Artificial intelligence basics

- Application integration: AI assistants inside office, help desk, and code tools; policy sets appropriate use and forbids plagiarism (passing AI output off as one's own)
- Limitations: bias from training data, hallucinations (confident fabrications), accuracy that must be verified
- Private versus public AI: public services may store and learn from what you type; never paste confidential data; data security, data source, and data privacy decide which tool is allowed